

Meeting 7

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Date:	8 July 2025
Subject:	How Stable are Baselines: analysis on the Consistency of Reinforcement Learning Baselines
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Project term:	February 2025 – August, 2025
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Outcome of previous meeting

- Extend to use TorchRL (instead of Tianshou) to avoid cherry-picking baseline frameworks
- Discussed introduction figures to convey the intention of our research
- Continue writing and analyzing the frameworks (prioritizing Stable Baselines 3 and CleanRL)

Results during the last weeks

- Implementations of algorithms mainly differ in terms of 'engineering tricks' for stability, which affects the methodologies used.
 - Some hyperparameters within one framework may be constants in another framework and vice versa, such as exploration noise within TorchRLs TD3 implementation.
 - Some constants differ, not just within engineering tricks (such as orthogonal initialization in PPO), but also within formulas, such as CleanRLs Q -value loss implementation in SAC, which is doubled.
 - Some methods are differently implemented, such as enforcing action bounds (either clipping or via tanh and scaling). This results in (slightly) different action distributions given a μ_θ and σ_θ , retrieved from a double-headed neural network with parameters θ .
 - Stable Baselines 3's off policy algorithms are trained on scaled actions, i.e. $a \in [-1, 1]$.
 - Truncation is handled differently across frameworks and across algorithms.
- There are some inconsistencies among algorithms seen already for PPO and TD3. It could mean that some environments are unstable to train on, or that there is an inconsistent behaviour within the algorithms.

Plans for next weeks

- Finalize the experiments, and link the analysis to possibly explain the inconsistencies found.
- First draft → End of July would be a safe date given aforementioned planning.